

Exploratory Data Analysis (EDA) Summary Report

1. Introduction

Purpose of the Report:

This report presents an Exploratory Data Analysis (EDA) of Geldium Finance's customer delinquency dataset. The purpose of the analysis is to assess the dataset's quality, completeness, and suitability for predictive modelling. By examining customer financial and behavioural attributes, the analysis aims to identify data quality issues, uncover meaningful patterns, and highlight potential indicators of credit card delinquency.

Goals of the Report:

The key goals of this report are to:

- Evaluate the structure, completeness, and consistency of the dataset.
- Identify missing values, anomalies, and data quality issues that may affect predictive modelling.
- Determine appropriate strategies for handling missing or inconsistent data.
- Explore relationships between customer attributes and delinquency outcomes.
- Identify key risk factors that may contribute to missed payments and delinquent accounts.
- Provide actionable insights that support the development of an AI-powered delinquency prediction and intervention framework.
- Ensure the dataset is prepared for future machine learning models while maintaining fairness, transparency, and data integrity.

2. Dataset Overview

The dataset was generally well-structured and suitable for analysis. However, several quality issues were identified:

- No duplicate Customer_ID records detected.
- Missing values present in Income (39), Loan_Balance (29), and Credit_Score (2).
- Employment_Status contains inconsistent labels (EMP, Employed, employed).
- 4 Credit_Utilization values exceed 100% (greater than 1.0).
- Class imbalance between delinquent and non-delinquent customers.

Key dataset attributes:

- Number of records: 500

- **Key variables:** Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Employment_Status, Account_Tenure, Credit_Card_Type, Location, and six months of payment history (Month_1–Month_6). These variables capture customer demographics, financial capacity, credit behavior, repayment history, and account characteristics, making them highly relevant for predicting delinquency risk.

- **Data types:**

Numerical Variables (8): Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure.

Categorical Variables (10): Customer_ID, Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6.

Binary Target Variable (1): Delinquent_Account (0 = Non-Delinquent, 1 = Delinquent).

3. Missing Data Analysis

Key missing data findings:

- **Variables with missing values:** Income (39 records-7.8%), Loan_Balance (29 records – 5.8%) and Credit_Score (2 records – 0.4%)

- **Missing data treatment:**

- Missing data treatment: Imputation was selected rather than deletion to preserve valuable customer information and maintain the dataset size. Median imputation was applied to Income and Loan_Balance due to potential outliers in financial data, while mean imputation was recommended for Credit_Score because of the very small number of missing observations.
- No synthetic data generation was required because the percentage of missing values was relatively low and could be addressed effectively through standard imputation techniques.

4. Key Findings and Risk Indicators

Key findings:

- **Correlations observed between key variables:**

- Higher numbers of missed payments are associated with increased delinquency risk.
- Customers with high credit utilization and high debt-to-income ratios appear more likely to become delinquent.
- Employment status may influence repayment behavior, with unemployed customers exhibiting higher delinquency rates.

- Income, debt burden, and repayment history collectively provide important indicators of customer credit risk.

- Unexpected anomalies:

- Missing values were identified in Income, Loan_Balance, and Credit_Score.
- Inconsistent Employment_Status labels (EMP, employed, Employed) require standardization.
- Several Credit_Utilization values exceed 100%, warranting further investigation.
- The dataset exhibits class imbalance, with delinquent accounts representing only 16% of observations.

5. AI & GenAI Usage

GenAI was used to complement the exploratory data analysis process by providing guidance on data preparation techniques and validating analytical observations. The dataset review, identification of missing values, anomaly detection, and interpretation of business implications were performed through direct analysis of the customer data. AI-generated recommendations were used to support decision-making regarding imputation strategies, risk factor assessment, and report refinement.

Example AI prompts used:

- "Summarize key patterns, outliers, and missing values in the dataset."
- "Suggest an imputation strategy for missing income values based on data characteristics and best practices."
- "Identify variables most likely to predict delinquency and explain their relevance."

6. Conclusion & Next Steps

The dataset is suitable for predictive modeling after data cleaning and preprocessing. Missing financial variables should be imputed, employment categories standardized, and anomalous credit utilization values reviewed. The most important potential delinquency predictors identified during EDA were Missed Payments, Credit Utilization, and Debt-to-Income Ratio.

Recommended next steps include:

1. Complete data preprocessing.
2. Engineer delinquency-related features.
3. Address class imbalance.
4. Develop and evaluate predictive models.
5. Build an intervention recommendation framework for the Collections team.